

System of Identical Particles

Books: ① Reif, Statistical Physics
② Chattopadhyay - Rakshit, Quantum Mechanics, Statistical Mechanics and Solid State Physics

Consider N identical particles in a volume V .

Let Q_i denote the coordinates of the i^{th} particle and s_i denote its state (e.g. its momentum, spin, etc.)

The joint wavefunction of the gas is denoted by $\psi_{\{s_1, s_2, \dots, s_N\}}(Q_1, Q_2, \dots, Q_N)$.

Classical Gas

- Distinguishable particles
- Any number of particles can be in the same state
- No restriction if two particles are exchanged

Energy of the state $r \rightarrow E_r$

Number of particles in state $r \rightarrow n_r$

States of the gas $\rightarrow R = \{n_r\}$

Total energy, $E_R = \sum_r n_r E_r$

Total number of particles $= N = \sum_r n_r$

Canonical partition function, $Z = \sum_R e^{-\beta E_R}$, where $\beta = \frac{1}{k_B T}$

Average number of particles in state $s = \bar{n}_s$

$$\bar{n}_s = \frac{\sum_R n_s e^{-\beta E_R}}{\sum_R e^{-\beta E_R}} = -\frac{1}{\beta Z} \frac{\partial Z}{\partial E_s} = -\frac{1}{\beta} \frac{\partial}{\partial E_s} \ln Z$$

$$\text{where } Z = \sum_R e^{-\beta E_R} = \sum_{n_1} \sum_{n_2} \dots \underbrace{\frac{N!}{n_1! n_2! \dots}}_{\substack{\sum_r n_r = N \\ \text{Number of ways of} \\ \text{putting } n_1 \text{ particles} \\ \text{in state 1, } n_2 \text{ in} \\ \text{state 2, } \dots}} e^{-\beta(E_1 n_1 + E_2 n_2 + \dots)}$$

$$\Rightarrow Z = \left(e^{-\beta E_1} + e^{-\beta E_2} + \dots \right)^N \quad \dots (1)$$

$$\Rightarrow \bar{n}_s = -\frac{1}{\beta Z} \frac{\partial Z}{\partial E_s} = -\frac{1}{\beta Z} N \left(e^{-\beta E_1} + e^{-\beta E_2} + \dots \right)^{N-1} \left(-\beta e^{-\beta E_s} \right)$$

$$\Rightarrow \boxed{\bar{n}_s = \frac{e^{-\beta E_s}}{\sum_r e^{-\beta E_r}} N}$$

Maxwell-Boltzmann distribution

Alternate derivation :

$$Z = \sum_{r_1} \sum_{r_2} \dots \sum_{r_N} e^{-\beta(\epsilon_{r_1} + \epsilon_{r_2} + \dots + \epsilon_{r_N})}$$
$$= \left(\sum_r e^{-\beta \epsilon_r} \right)^N \quad (\text{same as Eq. 1})$$

$r_i =$ state occupied by the i^{th} particle

Quantum Gas

There are two types of particles —

- (1) Particles with integer spins, which obey Bose-Einstein statistics
- (2) Particles with half-integer spins, which obey Fermi-Dirac statistics

} Spin-statistics theorem

These particles are called bosons and fermions, respectively.

* Half integer $\equiv$ Odd multiple of $1/2$

Exchange of two identical quantum particles

Let $\psi_{s_1 s_2}(\vec{x}_1, \vec{x}_2)$ denote the joint wavefunction of two particles in states s_1 and s_2 , at locations $\vec{x}_1$ and $\vec{x}_2$, respectively.

The joint wavefunction of two bosons must be symmetric under the exchange of the two bosons, i.e.,

for bosons, $\psi_{s_2 s_1}(\vec{x}_2, \vec{x}_1) = \psi_{s_1 s_2}(\vec{x}_1, \vec{x}_2)$

The joint wavefunction of two fermions must be antisymmetric under the exchange of the two fermions, i.e.,

for fermions, $\psi_{s_2 s_1}(\vec{x}_2, \vec{x}_1) = -\psi_{s_1 s_2}(\vec{x}_1, \vec{x}_2)$

Consequence: For fermions,

$$\psi_{ss}(\vec{x}, \vec{x}) = -\psi_{ss}(\vec{x}, \vec{x}) \Rightarrow \psi_{ss}(\vec{x}, \vec{x}) = 0$$

This is the Pauli's exclusion principle — Two fermions cannot be in the same state.

Example: Two identical particles, A and B, occupying 3 states

Classical
(Maxwell-Boltzmann)

state 1	AB	-	-	-	B	A	-	A	B
state 2	-	AB	-	A	-	B	B	-	A
state 3	-	-	AB	B	A	-	A	B	-

9 micro states

Bosons

state 1	AA	-	-	-	A	A
state 2	-	AA	-	A	-	A
state 3	-	-	AA	A	A	-

6 micro states

Fermions

state 1	-	A	A
state 2	A	-	A
state 3	A	A	-

3 micro states

Photon statistics (Bosons with unspecified number of total particles)

$$\bar{n}_s = \frac{\sum_{n_1, n_2, \dots} n_s e^{-\beta(n_1 \epsilon_1 + n_2 \epsilon_2 + \dots + n_s \epsilon_s + \dots)}}{\sum_{n_1, n_2, \dots} e^{-\beta(n_1 \epsilon_1 + n_2 \epsilon_2 + \dots + n_s \epsilon_s + \dots)}} \quad n_r \in \{0, 1, 2, \dots\}$$

$$= \frac{\sum_{n_s} n_s e^{-\beta n_s \epsilon_s}}{\sum_{n_s} e^{-\beta n_s \epsilon_s}} = -\frac{1}{\beta} \frac{\partial}{\partial \epsilon_s} \ln \sum_n e^{-\beta n \epsilon_s}$$

$$\sum_{n=0}^{\infty} e^{-\beta n \epsilon_s} = \frac{1}{1 - e^{-\beta \epsilon_s}}$$

$$\Rightarrow \bar{n}_s = \frac{1}{\beta} \frac{\partial}{\partial \epsilon_s} \ln(1 - e^{-\beta \epsilon_s}) = \frac{1}{\beta} \frac{\beta e^{-\beta \epsilon_s}}{1 - e^{-\beta \epsilon_s}} = \frac{1}{e^{\beta \epsilon_s} - 1}$$

$$\Rightarrow \boxed{\bar{n}_s = \frac{1}{e^{\beta \epsilon_s} - 1}} \quad \leftarrow \text{Planck distribution}$$